# Tenant-ive Plans: Predicting Residential Tenant Move-Out Risk

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## I. Updated Problem Statement

The central research question is whether a nonlinear feed-forward neural network (FFNN) can provide better generalization than a simpler logistic regression model when predicting weekly tenant move-out risk. The objective is to determine whether tenant and household characteristics can estimate the probability that a household moves out during a particular week. We treat the project as a discrete-time hazard classification problem: for every observed resident-week, the model predicts the probability that the household moves out during that week, conditional on still being present at the beginning of the week. Because the number of independent households is small compared with the number of weekly observations, the implementation emphasizes household-level splitting, leakage prevention, probability-based evaluation, and direct comparison between linear and nonlinear model capacity.

## II. Dataset and Preprocessing

The project uses anonymized residential property-management data from two properties. After validation, the dataset contains 455 resident records and 6,731 weekly observations. The modeling population contains 444 households, of which 125 experience a recorded move-out event; the remaining households are right-censored within the available observation period. Each modeling row corresponds to one resident during one observed tenancy week. The target, `churned_this_week`, equals one only during the final recorded week when a move-out occurs and zero otherwise.

The preprocessing script validates that household IDs are unique, no household-week appears twice, each household has at most one move-out, any move-out occurs on the final observed week, and row counts match the provider documentation. Eleven resident records have no corresponding weekly-panel observations and are therefore excluded from the modeling population.

Features are limited to information known at the start of the week being predicted. Tenant summaries such as total amount paid, late-payment counts, and total tenure are excluded to prevent leakage. Instead, the weekly outstanding-balance and delinquency variables are shifted by one week, producing `outstanding_lag1` and `delinquent_lag1`. Protected attributes identified in the provider documentation are excluded; number of key fobs is used as a household-size proxy. The final feature set contains nine numeric inputs—week index, the two lagged billing features, estimated monthly income, rent-to-income ratio, commute time, prior-home distance, fob count, and pet count—plus the categorical site/property feature.

Rent-to-income ratios below 0 or above 1 are set to missing. Missing numeric values are median-imputed during model fitting, so the imputation statistics are learned only from training data rather than the full dataset. Numeric variables are standardized, and site/property is one-hot encoded.

Because each household spans multiple weeks, all weeks from one household remain in the same data split. Repeated stratified 5-fold cross-validation divides the 444 households into folds of 88 or 89 households, with exactly 25 move-outs in each fold. Each fold serves as the held-out set while the other four folds are used for training, and this process is repeated five times. A fixed random seed makes the splits reproducible. The preprocessing pipeline also creates bootstrap samples for later robustness analysis.

## III. Baseline and Proposed Neural Model

The baseline is logistic regression implemented as a single PyTorch linear layer and trained with `BCEWithLogitsLoss`. Because move-out weeks are rare relative to stay-weeks, the loss is positively class-weighted using the ratio of stay-weeks to move-out-weeks in the training data. Adam is used with an initial learning rate of 0.05, a batch size of 64, and 200 epochs; L2 weight decay remains a tunable setting.

Both models use the same nine numeric inputs and one categorical site/property feature. The initial FFNN uses the same preprocessing, household split, target, class weighting, and evaluation metrics as the baseline, but adds two nonlinear hidden layers containing 16 and 8 neurons. Each hidden layer uses ReLU activation and 0.20 dropout. A final one-neuron layer produces the weekly move-out logit and is trained with `BCEWithLogitsLoss`. Keeping the data and evaluation procedure fixed allows the experiment to isolate the effect of nonlinear model capacity.

TABLE I
PRELIMINARY WEEKLY RESULTS, REPEAT 0 / FOLD 0

| Model | Split | Brier ↓ | Log Loss ↓ | AUC ↑ |
|---|---|---|---|---|
| Logistic | Train | 0.3298 | 0.8885 | 0.6246 |
| Logistic | Eval | 0.3258 | 0.8636 | 0.5475 |
| FFNN 16–8 | Train | 0.1755 | 0.5053 | 0.8602 |
| FFNN 16–8 | Eval | 0.2075 | 0.5793 | 0.5540 |

TABLE II
HOUSEHOLD-LEVEL CUMULATIVE MOVE-OUT RESULTS (EVAL FOLD)

| Model | Actual | Expected | Pred. | TP | FP | TN | FN | Precision | Recall | F1 |
|---|---|---|---|---|---|---|---|---|---|---|
| Logistic | 25 | 87.84 | 89 | 25 | 64 | 0 | 0 | 0.2809 | 1.0000 | 0.4386 |
| FFNN 16–8 | 25 | 76.67 | 79 | 22 | 57 | 7 | 3 | 0.2785 | 0.8800 | 0.4231 |

## IV. PRELIMINARY EXPERIMENT AND RESULTS

A first controlled comparison uses repeat 0, fold 0 as the held-out evaluation fold. The training portion contains 5,523 resident-weeks from 355 households, including 100 move-out events. The held-out portion contains 1,208 resident-weeks from 89 unseen households, including 25 move-outs. Performance is reported using Brier score, log loss, and ROC-AUC. Brier score measures squared error in predicted probabilities [1]; log loss penalizes confident incorrect probability estimates [2]; and ROC-AUC measures the ability to rank positive examples above negative examples across thresholds [3].

On the held-out fold, the FFNN performs better than logistic regression on all three weekly metrics. However, the FFNN shows substantial overfitting: training ROC-AUC reaches 0.8602 while evaluation ROC-AUC is 0.5540. Its training loss falls from approximately 1.36 to 0.95 while evaluation loss rises from approximately 1.46 to 1.73. The network is therefore becoming increasingly effective at fitting the training households without obtaining a corresponding improvement on unseen residents.

A household-level diagnostic converts each resident's weekly hazard probabilities into cumulative move-out risk using a 0.50 threshold. Table II shows that logistic regression predicts all 89 evaluation households will move out, resulting in 25 true positives, 64 false positives, and no true negatives. The FFNN is slightly more selective, predicting 79 move-outs and correctly identifying 22 of the 25 actual move-outs, but it still produces 57 false positives. Both models therefore substantially overestimate cumulative move-out risk.

A likely contributor is the positive-class weighting used to compensate for severe class imbalance. Only 100 of the 5,523 training resident-weeks contain move-out events, so positive observations receive much greater weight during training. This prevents the model from ignoring move-outs but may inflate weekly probabilities, which then compound across many weeks. Households are also observed for different lengths of time, so longer tenancies provide more opportunities for cumulative risk to increase.

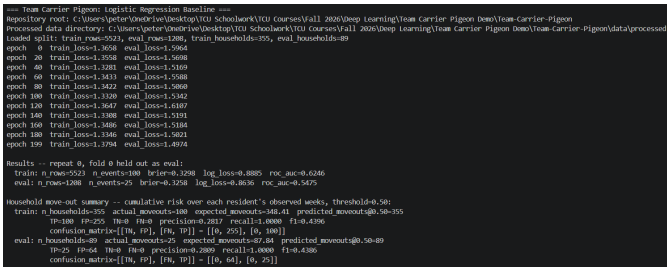



Fig. 1. Baseline logistic-regression output for repeat 0, fold 0.

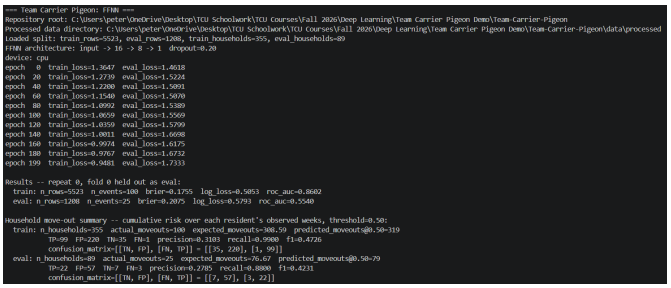



Fig. 2. FFNN output for repeat 0, fold 0.

## V. CURRENT DIFFICULTIES, LIMITATIONS, AND REMAINING WORK

The main limitation is the small number of independent households. Although the modeling table contains 6,731 weekly observations, those rows come from only 444 households, and repeated weekly measurements from one household are not independent. A second challenge is severe class imbalance: only 125 households have observed move-outs. In addition, some commute data is estimated rather than directly observed, and some income-related values require cleanup because of inconsistent units in the source data. Additional data has therefore been requested from the client.

The next stage will systematically test neural-network capacity and regularization. Planned architectures include single hidden layers with 4 or 8 neurons and deeper configurations such as 16–8, 32–16, and 64–32–16. Dropout strength, L2 weight decay, learning rate, and loss function will also be varied. Early stopping will use validation households drawn only from the training portion so that the outer held-out fold remains untouched during model selection. Promising configurations will be evaluated across all 25 repeated cross-validation folds, with final comparisons emphasizing fixed 26- and 52-week horizons using Brier score, calibration, log loss, and ROC-AUC.

## CODE AVAILABILITY

Project repository: https://github.com/pvo18/Team-Carrier-Pigeon

## REFERENCES

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[2] T. Gneiting and A. E. Raftery, “Strictly proper scoring rules, prediction, and estimation,” *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007, doi: 10.1198/016214506000001437.
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